🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 10 · STANDARD 6.GR.4

Surface Area of Prisms

Lesson 10-4 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the surface area of rectangular and triangular prisms.

Language Objective: I can explain my work using the words surface area, rectangular prism, triangular prism, and face.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Surface area | Rectangular prism | Triangular prism | Face | Net | Lateral face

👆 *Tap any word to see what it means and a picture.*

1. The total area of all the faces of a solid is its ____________.
2. A solid with six rectangular faces is a ____________.
3. A solid with two triangular faces and three rectangular faces is a ____________.
4. A flat surface of a solid figure is a ____________.
5. A flat pattern that folds into a solid is a ____________.
6. A side face of a prism that is not a base is a ____________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

For the triangular prism (base 6, height 4, length 10, slant sides 5), how did you account for the triangular bases AND the rectangular lateral faces?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Surface area** — The total area of all the flat sides of a solid.
  Área de superficie — El área total de todos los lados planos de un sólido.
- [ ] **Triangular prism** — A solid with two triangle ends and three flat rectangle sides.
  Prisma triangular — Un sólido con dos extremos triangulares y tres lados rectangulares.
- [ ] **Face** — One flat side of a solid shape.
  Cara — Un lado plano de una figura sólida.
- [ ] **Net** — A flat shape that folds up into a solid.
  Plantilla (desarrollo plano) — Una figura plana que se dobla y forma un sólido.
- [ ] **Lateral face** — The side faces of a solid, not the top or bottom.
  Cara lateral — Los lados de un sólido, no la parte de arriba ni la de abajo.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The two triangular bases together are $2 \times (\frac{1}{2} \times 6 \times 4) =$ ____.

Las dos bases triangulares juntas son $2 \times (\frac{1}{2} \times 6 \times 4) =$ ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: For both, you add the areas of all the faces, but the shapes and number of faces differ.

Evidence: Students explain both methods add up all face areas, but a rectangular prism has 6 rectangular faces while a triangular prism has 5 faces (2 triangles + 3 rectangles).

Reasoning: This evidence connects the definition of surface area to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The two triangular bases together are 2 × (½ × 6 × 4) = ____.**
  Las dos bases triangulares juntas son 2 × (½ × 6 × 4) = ____.
- **The three lateral faces add to ____, so the total surface area is ____.**
  Las tres caras laterales suman ____, así que el área total de superficie es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

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4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Surface area
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Triangular prism
4. **Notes 4:** Face
5. **Notes 5:** Net
6. **Notes 6:** Lateral face
7. **Watch for this mistake:** *A common mistake in Surface Area of Prisms is confusing surface area with volume — multiplying l × w × h (which finds the space INSIDE the prism, in cubic units) instead of adding the areas of all the faces (SA = 2lw + 2lh + 2wh, in square units). A second common slip is forgetting to double one face pair, like writing 2(l×w) + 2(l×h) + (w×h) instead of + 2(w×h) — always count all 6 faces (or all 5 for a triangular prism) before you submit.*
8. **Try It 1:** A. 148 cm^2 — *SA = 2(6×5) + 2(6×4) + 2(5×4) = 60 + 48 + 40 = 148* cm^2.
9. **Try It 2:** A. 288 in^2 — *Two triangle ends: 2 × (½ × 8 × 6) = 48* in^2. *Three rectangles: (8 + 6 + 10) × 10 = 24 × 10 = 240* in^2. *SA = 48 + 240 = 288* in^2.